

$$\underbrace{1 \dots \dots 1}_n(2) = \underbrace{10 \dots \dots 0}_n(2) - 1(2) = 2^n - 1$$

Example 1

n=8 bits

$$X = 47 = 32 + 8 + 4 + 2 + 1 = 2^5 + 2^3 + 2^2 + 2^1 + 2^0 = 101111(2)$$

$$Y = 96 = 64 + 32 = 2^6 + 2^5 = 1100000(2)$$

positions	S	7	6	5	4	3	2	1	0
$[47]_{\text{dir}} = [47]_{\text{inv}} = [47]_{\text{compl}} =$	0	0	1	0	1	1	1	1	1
$[-47]_{\text{dir}} =$	1	0	1	0	1	1	1	1	1
$[-47]_{\text{inv}} =$	1	1	0	1	0	0	0	0	0
$[-47]_{\text{compl}} =$	1	1	0	1	0	0	0	0	1

positions	S	7	6	5	4	3	2	1	0
$[96]_{\text{dir}} = [96]_{\text{inv}} = [96]_{\text{compl}} =$	0	1	1	0	0	0	0	0	0
$[-96]_{\text{dir}} =$	1	1	1	0	0	0	0	0	0
$[-96]_{\text{inv}} =$	1	0	0	1	1	1	1	1	1
$[-96]_{\text{compl}} =$	1	0	1	0	0	0	0	0	0

$$[47+96]_{\text{compl}} = [47]_{\text{compl}} \oplus [96]_{\text{compl}}$$

		S									Overflow because the operands are positive numbers and the result is negative (the sign bit is 1)
[47] _{compl} =		0	0	1	0	1	1	1	1	⊕	
[96] _{compl} =		0	1	1	0	0	0	0	0		
		1	0	0	0	1	1	1	1		

$$[96-47]_{\text{compl}} = [96]_{\text{compl}} \oplus [-47]_{\text{compl}}$$

		S									Correct result $2^0 + 2^4 + 2^5 = 1 + 16 + 32 = 49$
$[96]_{\text{compl}} =$		0	1	1	0	0	0	0	0	$\oplus$	
$[-47]_{\text{compl}} =$		1	1	0	1	0	0	0	1		
$[49]_{\text{compl}}$	1	0	0	1	1	0	0	0	1		

$$[47-96]_{\text{compl}} = [47]_{\text{compl}} \oplus [-96]_{\text{compl}}$$

		S									Correct result
[47] _{compl} =		0	0	1	0	1	1	1	1	⊕	-(1+2 ⁴ +2 ⁵)=-(1+16+32)=-49

$[-96]_{\text{compl}} =$		1	0	1	0	0	0	0	0		
$[-49]_{\text{compl}}$		1	1	0	0	1	1	1	1		
		0	0	1	1	0	0	0	1		complement

Subunitary numbers – codes and operations in complementary code

Example 2

n=8 bits

$$X = 0,31 = 0,0100111(2)$$

$$0,31 * 2 = 0,62$$

$$0,62 * 2 = 1,24$$

$$0,24 * 2 = 0,48$$

$$0,48 * 2 = 0,96$$

$$0,96 * 2 = 1,92$$

$$0,92 * 2 = 1,84$$

$$0,84 * 2 = 1,68$$

$$Y = 0,73 = 0,565(8) = 0,101110101(2)$$

$$0,73 * 8 = 5,84$$

$$0,84 * 8 = 6,72$$

$$0,72 * 8 = 5,76$$

positions	S	7	6	5	4	3	2	1	0
$[0,31]_{\text{dir}} = [0,31]_{\text{inv}} = [0,31]_{\text{compl}} =$	0	0	1	0	0	1	1	1	
$[-0,31]_{\text{dir}} =$	1	0	1	0	0	1	1	1	
$[-0,31]_{\text{inv}} =$	1	1	0	1	1	0	0	0	
$[-0,31]_{\text{compl}} =$	1	1	0	1	1	0	0	1	

positions	S	7	6	5	4	3	2	1	0
$[0,73]_{\text{dir}} = [0,73]_{\text{inv}} = [0,73]_{\text{compl}} =$	0	1	0	1	1	1	0	1	
$[-0,73]_{\text{dir}} =$	1	1	0	1	1	1	0	1	
$[-0,73]_{\text{inv}} =$	1	0	1	0	0	0	1	0	
$[-0,73]_{\text{compl}} =$	1	0	1	0	0	0	1	1	

$$[0,31+0,73]_{\text{compl}} = [0,31]_{\text{compl}} \oplus [0,73]_{\text{compl}}$$

	S									
$[0,31]_{\text{compl}} =$	0	0	1	0	0	1	1	1	$\oplus$	Overflow because the operands are positive and the result is negative (the sign bit is 1)
$[0,73]_{\text{compl}} =$	0	1	0	1	1	1	0	1		
	1	0	0	0	0	1	0	0		

$$[0,31]_{\text{compl}} = [0,31]_{\text{compl}} \oplus [-0,31]_{\text{compl}}$$

		S ,									
$[0,31]_{\text{compl}} =$		0	0	1	0	0	1	1	1	$\oplus$	Correct result: 0
$[-0,31]_{\text{compl}} =$		1	1	0	1	1	0	0	1		
$[0]_{\text{compl}}$	1	0	0	0	0	0	0	0	0		

$$[0,73]_{\text{compl}} = [0,73]_{\text{compl}} \oplus [-0,31]_{\text{compl}}$$

		S ,									
$[0,73]_{\text{compl}}$		0	1	0	1	1	1	0	1	$\oplus$	Correct result $2^{(-2)}+2^{(-3)}+2^{(-5)}+2^{(-6)}=\frac{1}{4}+\frac{1}{8}+\frac{1}{32}+\frac{1}{64}=0,25+0,125+0,03125+0,015625=0,421875$
$[-0,31]_{\text{compl}} =$		1	1	0	1	1	0	0	1		
	1	0	0	1	1	0	1	1	0		

Example 3: Represent in fixed-point notation, on 32 bits, I=14 bits, the number 3456,73

$$3456,73 = 110\ 110\ 000\ 000, 101\ 110\ 101\ 110\ 000\ 101(2)$$

$$3456 = 6600(8) = 110\ 110\ 000\ 000(2)$$

$$3456:8=432\ r\ 0$$

$$432:8 = 54\ r\ 0$$

$$54:8 = 6\ r\ 6$$

$$6:8 = 0\ r\ 6$$

$$0,73 = 0,565605(8) = 0,101\ 110\ 101\ 110\ 000\ 101(2)$$

$$0,73 \cdot 8 = 5,84$$

$$0,84 \cdot 8 = 6,72$$

$$0,72 \cdot 8 = 5,76$$

$$0,76 \cdot 8 = 6,08$$

$$0,08 \cdot 8 = 0,64$$

$$0,64 \cdot 8 = 5,12$$

$$1011(2) = 2^0 + 2^1 + 2^3 = 13 = D(16)$$

S	I=14 bits, integer part ->														,	<- F=17 bits, fractional part															
0	0	0	1	1	0	1	1	0	0	0	0	0	0	1	0	1	1	1	0	1	0	1	1	1	0	0	0	0	1	0	
1	B			0			1			7			5			C			2												

Example 4: Represent in floating point notation, single precision (SP), with mantissa<1, the number: 3456,73

$$3456,73 = 110\ 110\ 000\ 000, 101\ 110\ 101\ 110\ 000\ 101(2) = 0,110\ 110\ 000\ 000101\ 110\ 101\ 110\ 000\ 101(2) * 2^{12}$$

$$:c = 127 + 12 = 139 = 128 + 8 + 2 + 1 = 2^7 + 2^3 + 2^1 + 2^0 = 10001011(2)$$

S	c=e+127 (8bits) -> , <- mantissa (23bits)																											
0	1	0	0	0	1	0	1	1	1	1	0	1	1	0	0	0	0	0	0	1	0	1	1	1	0	1	1	1
4	5				E				C				0				5				D				7			

Example 5: Find the real number X having C504A800 its fixed-point representation on 32 bits with I=17 bits.

S	I=17 bits, integer part																	->		<-	F=14 bits, fractional part												
1	1	0	0	0	1	0	1	0	0	0	0	0	1	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0			
C					5				0				4				A				8				0				0				

$$X = -10001010000010010,10100000000000(2) = -(2^{17} + 2^{13} + 2^{11} + 2^4 + 2^1 + 2^{-1} + 2^{-3}) = -(131072 + 8192 + 2048 + 16 + 2 + 0,5 + 0,125) = -141330,625$$

X = -141330,625

Example 6: Find the real number X having C504A800 as its floating-point representation, SP, m>1.

S	c=e+127 (8bits) -> , <- mantissa (23bits)																														
1	1	0	0	0	1	0	1	0	0	0	0	0	1	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0	0		
C				5				0				4				A				8				0				0			

$$c = 10001010(2) = 2^1 + 2^3 + 2^7 = 2 + 8 + 128 = 138$$

$$e = 138 - 127 = 11$$

$$X = -1,000010010101(2) * 2^{11}$$

$$= -100001001010,1(2)$$

$$= -(2^1 + 2^3 + 2^6 + 2^{11} + 2^{-1}) = -(2 + 8 + 64 + 2048 + 0.5) = -(74 + 2048 + 0.5) =$$

$$= -2122.5$$